

1, Data cleaning and transformation:

- **Data types:** All columns verified - dates set to Date, numeric fields to Integer/Decimal, text columns confirmed
- Duplicates removed from the raw table to ensure transactional integrity before normalisation
- Add index-based primary key column for Dim_Pizza and Dim_Category tables, followed by joining these 2 tables with the fact table and deleting redundant variables in the fact table namely “pizza_id”, “name”, “size”, “ingredients”, “category” for clearance.
- **Normalisation into snowflake schema:**
 - **Fact_Pizza_Table:** order_id, date, pizza_id, quantity, price; redundant descriptive columns removed after merging
 - **Dim_Pizza:** pizza_id (PK), name, category, size, ingredients; duplicates removed on pizza_id
 - **Dim_Date:** Date (PK) with Month and Week day created to support time intelligence
 - **Dim_Category:** distinct category values acting as a hub for cross-table filtering
 - **Ingredient_Sales:** built by splitting comma-separated ingredients to rows, trimming whitespace, grouping by category and ingredient to produce Total_quantity_of_ingredients_sold
- **Derived columns:** Hours, Week day, Month (for time analysis); Order type (Single/Multi-item); size_sort_order helper column (S=1 through XXL=5) used to sort sizes logically via Sort by Column
- **Model:** One-to-many relationships from each Dim to Fact; unnecessary Fact fields hidden; 5 DAX measures created

2, Data Analysis and Visualisation:

- **Core KPIs:** Total Revenue \$817.9K | Total Quantity Sold 49,574 | Total Orders 21,350 | Avg Order Value \$38.31 | Avg Pizzas/Order 2.32.
- **Monthly trends:** Revenue peaked in July, October, and November; lowest in March and December. For instance, the total revenue on July 4th was expected to be around \$2.3k given the evidence of previous revenue of around \$1.6k and \$3k; however, the reality showed that the revenue on that day was \$3.9k, reaching the highest from Jan 2015. This is an anomaly as it exceeds the prediction; this peak revenue could result from the school holidays, wage adjustments, etc.
- **Day/hour analysis:** Friday is the busiest day (3,538 orders); Sunday is weakest (2,624) - a 35% gap; 12:00 PM is the peak hour (lunch rush); secondary peak at 18:00-19:00; quietest periods are 14:00-15:00 and post-21:00
- **Category rankings:** Classic leads both quantity (14,888 units, \$220.6K revenue); Veggie ranks last in revenue (\$193.7K)
- **Top 5 by qty:** Classic Deluxe (2,453), BBQ Chicken (2,432), Hawaiian (2,422), Pepperoni (2,418), Thai Chicken (2,371)
- **Top 5 by revenue:** Thai Chicken (\$43.4K), BBQ Chicken (\$42.8K), California Chicken (\$41.4K), Classic Deluxe (\$38.2K), Spicy Italian (\$34.8K). Especially, Thai Chicken ranks 5th in qty but 1st in revenue, showing premium pricing power
- **Bottom 5 by qty:** Brie Carre (490), Mediterranean (934), Calabrese (937), Spinach Supreme (950), Soppressata (961). Brie Carre shows a 5× gap versus the top seller
- **Bottom 5 by revenue:** Brie Carre (\$11.6K), Green Garden (\$14.0K), Spinach Supreme (\$15.3K), Mediterranean (\$15.4K), Spinach Pesto (\$15.6K)
- **Size distribution:** Large (L) dominates across all categories (~19K units); Medium ~16K; Small ~14K; XL minimal (~1K); XXL near zero, indicating customers strongly prefer mid-to-large sizes
- **Ingredients:** Garlic is most popular, appearing in all top-selling pizzas; Brie Carre Cheese and Caramelized Onions are least popular, featuring only in bottom performers; negative price-volume correlation exists across categories
- **Order habits:** 62% multi-item vs 38% single-item orders

3, Recommendations:

- **Discontinue or reformulate Brie Carre Pizza:** 490 units sold and \$11.6K revenue represents a 5× gap to the top performer; its niche ingredients (Brie Carre Cheese, Caramelized Onions) have near-zero mass-market appeal; replacing it with a product built on proven ingredients (Garlic, Mozzarella, Tomato Sauce) would directly improve category efficiency and simplify procurement

- **Launch targeted seasonal promotions in March and December:** both months underperform compared to the other months, maybe for example, because of the variety of food options and shopping habits during Christmas and Boxing Day in December. Therefore, promotions such as Christmas discount program could be introduced to attract customers
- **Convert single-item orders through upselling at checkout:** with ~8,100 single-item transactions representing 38% of all orders, introducing a 'add a second pizza at a discount' prompt is one of the highest-ROI short-term lever available